

Project Structure

development folder

database.py --defined a py file to connect to our database

models.py --define our tables to classes so that python can easily consume

test.ipynb --check how to consume our data

Data Structure

1. properties Table (The Parent)

id: UUID [Primary Key] (Auto-generates via uuid4)

address: VARCHAR(255) [Not Null]

city: VARCHAR(100) [Not Null] [Indexed for fast searching]

zip_code: VARCHAR(20) [Not Null]

property_type: VARCHAR(50) [Not Null]

bedrooms: INTEGER

bathrooms: INTEGER

square_meters: FLOAT

year_built: INTEGER

Relationships: Cascades deletes to listings, audits, and anomalies.

2. listings Table (Child of Properties)

id: UUID [Primary Key]

property_id: UUID [Foreign Key -> properties.id] [ON DELETE CASCADE]

asking_price: NUMERIC(15, 2) [Not Null]

status: VARCHAR(50) [Not Null] [Indexed]

listed_date: DATETIME (Timezone aware) [Default: Current server time]

3. audits Table (Child of Properties)

id: UUID [Primary Key]

property_id: UUID [Foreign Key -> properties.id] [ON DELETE CASCADE]

estimated_rental_income: NUMERIC(12, 2)

estimated_maintenance_costs: NUMERIC(12, 2)

gross_yield_percentage: FLOAT

calculated_at: DATETIME (Timezone aware) [Default: Current server time]

4. anomalies Table (Child of Properties)

id: UUID [Primary Key]

property_id: UUID [Foreign Key -> properties.id] [ON DELETE CASCADE]

flag_type: VARCHAR(100) [Not Null]

description: TEXT [Not Null]

severity: VARCHAR(20) [Not Null]

Useful git commands

important: work on your own branch.

```
git clone <repository_url>
```

2. Pull Latest Changes

```
git pull origin <branch_name>
```

3. Add Changes

```
git add .
```

(or specify files)

```
git add <file_name>
```

4. Commit Changes

```
git commit -m "commit message"
```

5. Push Changes

```
git push origin <branch_name>
```

Example:

```
git push origin main
```

6. Create Branch

```
git branch <branch_name>
```

7. Switch Branch

```
git checkout <branch_name>
```